DBMS-SQL QUERY IMPLEMENTATION
PROJECT TITLE-SHOPWITHNYKAA
WEEK -7

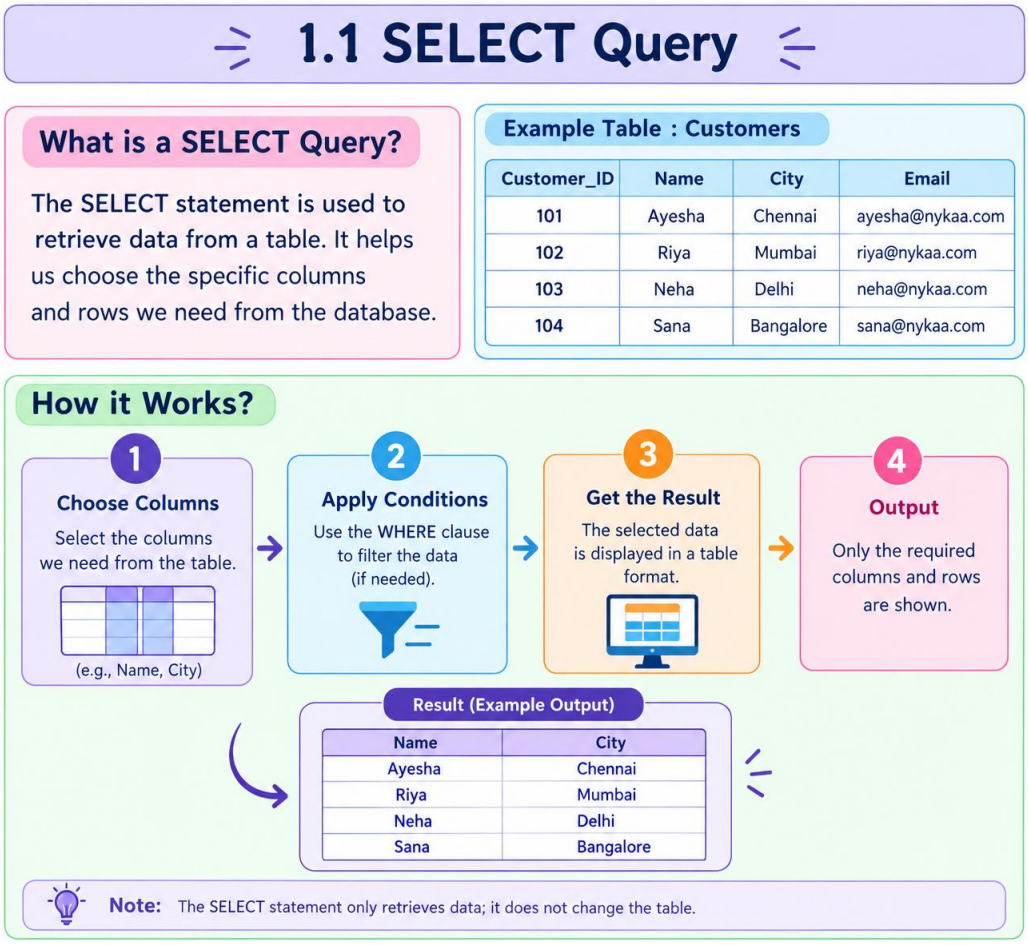
1. INTRODUCTION

SQL (Structured Query Language) is used to retrieve and manage data in a database. In the Nykaa E-Commerce Database, SQL queries are used to display customer details, search products by price and category, check product availability, and generate reports. Commands such as SELECT, WHERE, ORDER BY, and DISTINCT help filter and organize data efficiently.

2. OBJECTIVES

1. Perform SELECT, WHERE, ORDER BY, and DISTINCT queries.
2. Search products based on price, category, and availability.
3. Retrieve customer and product information.
4. Apply filtering conditions using AND, OR, IN, and BETWEEN.
5. Generate basic business reports for products, customers, and orders.

3. BASIC SQL QUERIES



4. Display all products

```
SQL> select * from product;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  BRAND_ID CATEGORY_ID      PRICE      STOCK
---------- ----------- ---------- ----------
DESCRIPTION
--------------------------------------------------------------------------------
SKIN_TYPE                                                  RATING EXPIRY_DA
-------------------------------------------------- ---------- ---------
      1001
Vitamin C Face Serum
       101         101        649         45

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  BRAND_ID CATEGORY_ID      PRICE      STOCK
---------- ----------- ---------- ----------
DESCRIPTION
--------------------------------------------------------------------------------
SKIN_TYPE                                                  RATING EXPIRY_DA
-------------------------------------------------- ---------- ---------
Vitamin C serum for glowing skin
All Skin Types                                            4.5 31-DEC-28


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  BRAND_ID CATEGORY_ID      PRICE      STOCK
---------- ----------- ---------- ----------
DESCRIPTION
--------------------------------------------------------------------------------
SKIN_TYPE                                                  RATING EXPIRY_DA
-------------------------------------------------- ---------- ---------
      1002
Matte Lipstick
       102         102        449         60

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  BRAND_ID CATEGORY_ID      PRICE      STOCK
---------- ----------- ---------- ----------
DESCRIPTION
--------------------------------------------------------------------------------
SKIN_TYPE                                                  RATING EXPIRY_DA
-------------------------------------------------- ---------- ---------
Long lasting matte lipstick
All Skin Types                                            4.5 31-JAN-29
```

5. Display selected product details

```
SQL> select product_id,product_name,price from product;
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
      1001
Vitamin C Face Serum
       649

      1002
Matte Lipstick
       449

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
      1003
Anti Dandruff Shampoo
       349

      1004
Floral Eau De Parfun
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       899

      1005
Body Lotion
       349

      1006
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
Lavender Shower Gel
       299

      1007
Red Nail Polish
       199

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
      1008
Lip Balm
       149
```

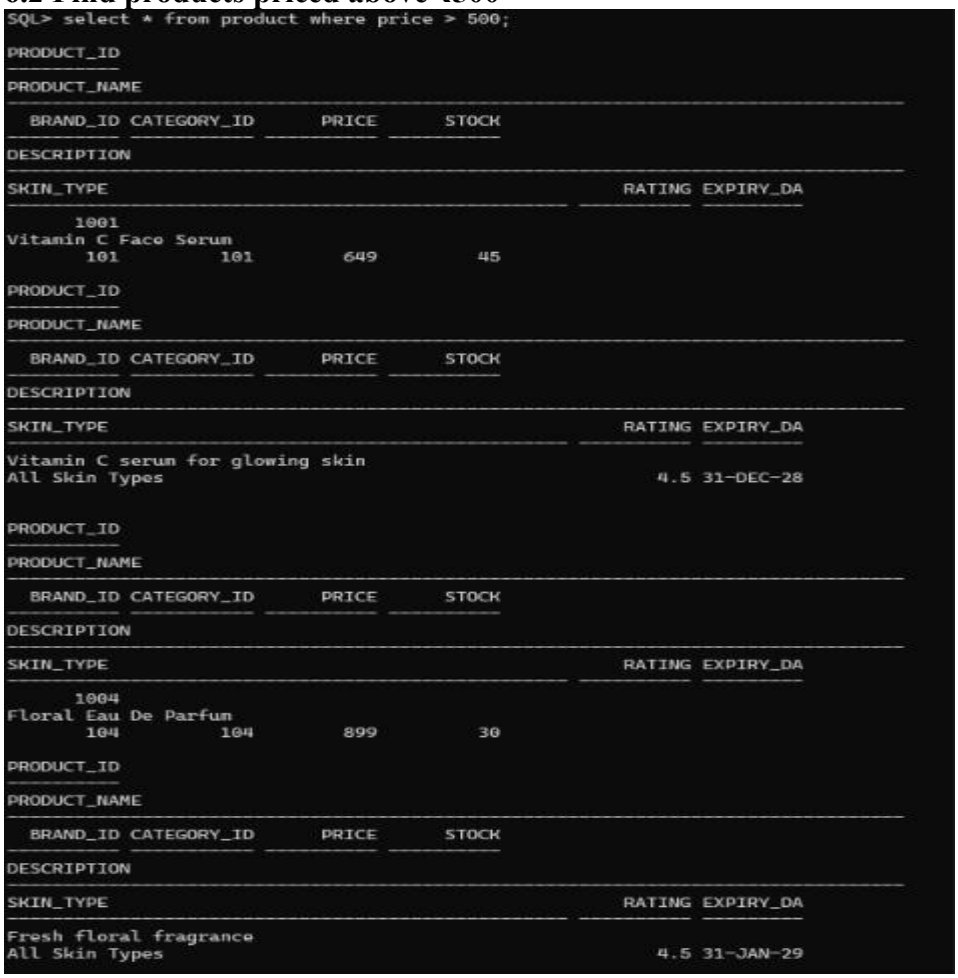
6. WHERE CLAUSE

The WHERE clause is used to retrieve records that satisfy a particular condition.

6.1 Find products priced below ₹500



6.2 Find products priced above ₹500



7. **ORDER BY**

The ORDER BY clause sorts the results in ascending or descending order.

7.1 Display products from lowest to highest price

```
SQL> select product_id,product_name,price from product order by price asc;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
      1008
Lip Balm
       149

      1007
Red Nail Polish
       199

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

      1006
Lavender Shower Gel
       299

      1003
Anti Dandruff Shampoo

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       349

      1005
Body Lotion
       349

      1002

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
Matte Lipstick
       449

      1001
Vitamin C Face Serum
       649

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
      1004
Floral Eau De Parfum
```

7.2 Display products from highest to lowest price

```
SQL> select product_id,product_name,price from product order by price desc;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
      1004
Floral Eau De Parfun
       899

      1001
Vitanin C Face Serun
       649

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

      1002
Matte Lipstick
       449

      1003
Anti Dandruff Shampoo

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       349

      1005
Body Lotion
       349

      1006

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
Lavender Shower Gel
       299

      1007
Red Nail Polish
       199

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
      1008
Lip Balm
       149
```

8. DISTINCT

The DISTINCT keyword eliminates duplicate values from the query results.

8.1 Display unique product categories

```
SQL> select distinct product_id,product_name,price from product;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
      1001
Vitamin C Face Serum
       649

      1002
Matte Lipstick
       449

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

      1003
Anti Dandruff Shampoo
       349

      1004
Floral Eau De Parfum

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       899

      1005
Body Lotion
       349

      1006

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
Lavender Shower Gel
       299

      1007
Red Nail Polish
       199


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
      1008
Lip Balm
       149
```

9. SEARCH PRODUCTS

9.1 Search products based on price

```
SQL> select  product_id,product_name,price from product where price between 100 and 400;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
      1003
Anti Dandruff Shampoo
       349

      1005
Body Lotion
       349

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

      1006
Lavender Shower Gel
       299

      1007
Red Nail Polish

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       199

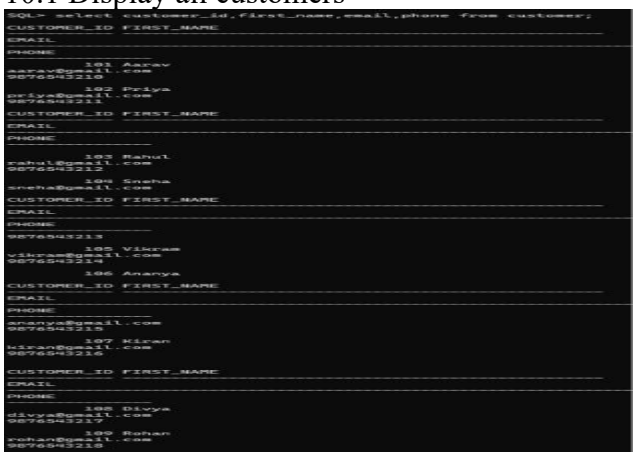
      1008
Lip Balm
       149
```

9.2 Search products based on category

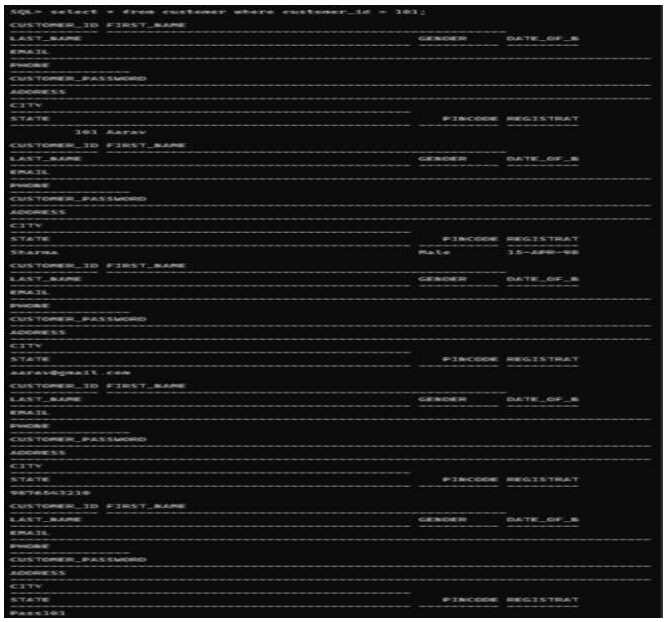


10. RETRIEVE CUSTOMER INFORMATION

10.1 Display all customers



11.Search for a particular customer



11. APPLY FILTERING CONDITIONS

SQL provides operators such as AND, OR, IN, and BETWEEN to filter data

11.1 AND Condition

```
SQL> select  product_id,product_name,price from product where price < 500 and category_id = 102;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
      1002
Matte Lipstick
       449
```

11.2 OR Condition

```
SQL> select  product_id,product_name,price from product where category_id = 101 or category_id = 103;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
      1001
Vitamin C Face Serum
       649

      1003
Anti Dandruff Shampoo
       349

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
```

11.3 IN Operator

```
SQL> select  product_id,product_name,price from product where category_id in (101,102,103);

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
      1001
Vitamin C Face Serum
       649

      1002
Matte Lipstick
       449

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

      1003
Anti Dandruff Shampoo
       349
```

11.4 BETWEEN Operator

```
SQL> select  product_id,product_name,price from product where price between 100 and 300;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
      1006
Lavender Shower Gel
       299

      1007
Red Nail Polish
       199

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

      1008
Lip Balm
       149
```

12. BASIC BUSINESS REPORTS

12.1 Product Price Report

```
SQL> select  product_id,product_name,price from product order by price desc;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
      1004
Floral Eau De Parfum
       899

      1001
Vitamin C Face Serum
       649

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

      1002
Matte Lipstick
       449

      1003
Anti Dandruff Shampoo

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       349

      1005
Body Lotion
       349

      1006

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
Lavender Shower Gel
       299

      1007
Red Nail Polish
       199


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
      1008
Lip Balm
```

12.2 Customer Order Report



```
SQL> select c.customer_id,c.first_name,o.order_id,o.order_date,o.total_amount from customer c join orders o on c.customer_id = o.customer_id order by o.order_date desc;

CUSTOMER_ID FIRST_NAME                                         ORDER_ID
----------- -------------------------------------------------- ---------
ORDER_DAT TOTAL_AMOUNT
--------- ------------
        105 Vikram                                                    5
10-SEP-26                950

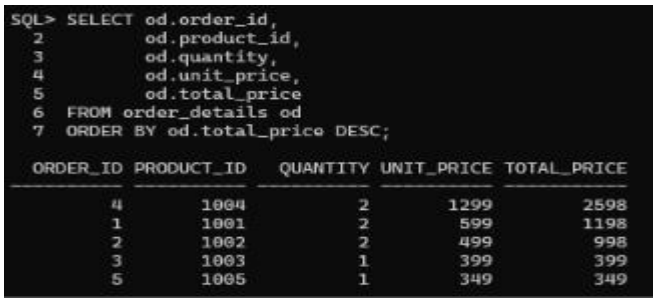
        104 Sneha                                                     4
08-SEP-26               3200

        103 Rahul                                                     3
05-SEP-26               1750

CUSTOMER_ID FIRST_NAME                                         ORDER_ID
----------- -------------------------------------------------- ---------
ORDER_DAT TOTAL_AMOUNT
--------- ------------
        102 Priya                                                     2
03-SEP-26               2300

        101 Aarav                                                     1
02-SEP-26               1600
```

12.3 Product Order Report



```
SQL> SELECT od.order_id,
  2         od.product_id,
  3         od.quantity,
  4         od.unit_price,
  5         od.total_price
  6  FROM order_details od
  7  ORDER BY od.total_price DESC;

  ORDER_ID PRODUCT_ID   QUANTITY UNIT_PRICE TOTAL_PRICE
---------- ---------- ---------- ---------- -----------
         4       1004          2       1299        2598
         1       1001          2        599        1198
         2       1002          2        499         998
         3       1003          1        399         399
         5       1005          1        349         349
```

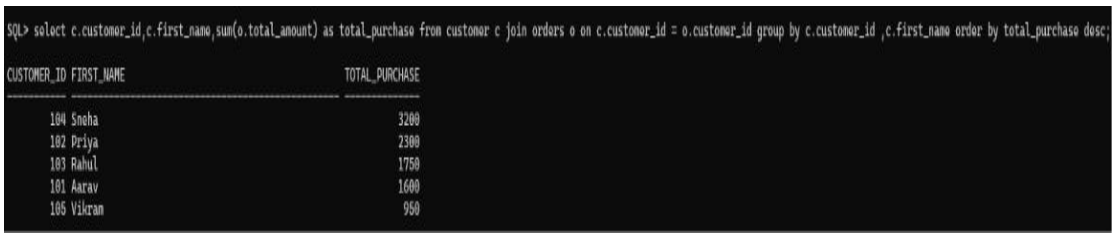
12.4 Customer-wise Total Orders



```
SQL> select c.customer_id,c.first_name,count(o.order_id) as total_orders from customer c join orders o on c.customer_id = o.customer_id group by c.customer_id,c.first_name order by total_orders desc;

CUSTOMER_ID FIRST_NAME                                    TOTAL_ORDERS
----------- -------------------------------------------------- ----------
        101 Aarav                                                   1
        102 Priya                                                   1
        105 Vikram                                                  1
        104 Sneha                                                   1
        103 Rahul                                                   1
```

12.5 Customer-wise Total Purchase Amount



```
SQL> select c.customer_id,c.first_name,sum(o.total_amount) as total_purchase from customer c join orders o on c.customer_id = o.customer_id group by c.customer_id ,c.first_name order by total_purchase desc;

CUSTOMER_ID FIRST_NAME                                    TOTAL_PURCHASE
----------- -------------------------------------------------- --------------
        104 Sneha                                                     3200
        102 Priya                                                     2300
        103 Rahul                                                     1750
        101 Aarav                                                     1600
        105 Vikram                                                     950
```

13. CONCLUSION

In this task, SQL queries were implemented in the Nykaa database to retrieve product and customer details. Products were filtered based on price and category. Data was sorted using SQL commands, and basic business reports were generated to analyze customer orders and purchases.